

15 September 2026 05:51

[10]

- CE

CA'I

- OR**

- $C'AE'$

Output: X [1 indicates bonus point awarded, 0 indicates bonus point not awarded for all cases]

[5]

- ## Cardinal notation

[3]

- even number of [2] zero's in a word

From <<https://teams.live.com/v2/>>

$$\begin{aligned} 0' &\rightarrow 1 \\ 1' &\rightarrow 0 \end{aligned}$$

[De Morgan's theorem]

$$= \underline{A' + A'B' + A'C'} + B'C' + \underline{A'B'C}$$

[properties of 1]

[distributive law]

Proof of distributive law:

$$A'(1+B'+C) + B'C$$

$$A' + B'C$$

$$X(c, A, \bar{E}, \bar{I})$$

$$= \begin{pmatrix} C+A+E+I \\ C+A+E+I' \\ C'+A+E+I \end{pmatrix} \begin{pmatrix} C+E+A+I' \\ C+A+E+I \\ C'+A+E+I \end{pmatrix} \begin{pmatrix} C+A+E'+I \\ C+A'+E'+I \\ C'+A+E+I' \end{pmatrix}$$

$$\rightarrow X(C, A, E, I) = \Pi(0, 1, 2, 3, 6, 7, 8, 12, 13).$$

$$X(C, A, E, I) = \sum (4, 5, 9, 10, 11, 14, 15)$$

even $\rightarrow$
odd $\leftarrow$
odd $\rightarrow$
even $\leftarrow$
odd

odd →
even →
odd →
even →

0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	0

Question 3

- (i) A shopping mall announces a special discount on all its products as a festival offer only to those who satisfy any one of the following conditions. [5]

- If he/she is an employee of the mall and has a service of more than 10 years. *ES*
- A regular customer of the mall whose age is less than 65 years and should not be an employee of the mall. *ER'C*
- If he/she is a senior citizen but not a regular customer of the mall. *R'C*

The inputs are :

INPUTS	
E	Employee of the mall
R	Regular customer of the mall
S	Service of the employee is more than 10 years
C	Senior citizen of 65 years or above

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: X - Denotes eligible for discount [1 indicates YES and 0 indicates NO in all cases]

Draw the truth table for the inputs and outputs given above and write the SOP expression for X (E, R, S, C). (Application)

INPUTS				OUTPUT
E	R	S	C	X
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

$$X(E, R, S, C) = \sum (1, 3, 4, 6, 9, 10, 11, 14, 15)$$

Question 4:

- a) A company wants to shortlist candidates for a leadership program based on the following criteria:

The candidate has at least 3 years of work experience and has led a team successfully. *WL*

OR

The candidate has at least 3 years of work experience and has received an award for leadership. *WR*

The inputs are:

Inputs	Description
W	Candidate has at least 3 years of work experience
L	Candidate has led a team successfully
R	Candidate has received an award for leadership

(In all cases, 1 indicates yes and 0 indicates no.)

Output Y - Denotes shortlisted (1 indicates shortlisted, 0 indicates not shortlisted).

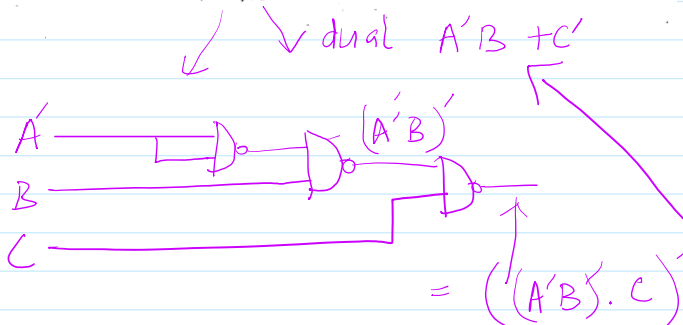
Draw the truth tables for the inputs and outputs given above and write the SOP expression for Y(W, L, R). [5]

- b) Simplify using Boolean laws [2]

$$X = (Y + Z(X.Y + X.Z))'$$

- c) Draw the logic gate diagram for the expression using NAND gate only. [3]

$$P = (A' + B) \cdot C'$$



INPUT			OUTPUT
W	L	R	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$Y(W, L, R) = \sum (5, 6, 7)$$

$$\begin{aligned}
 & X = (Y + Z(X.Y + X.Z))' \\
 & = X(Y + Z(XY)'(XZ))' \quad [\text{De-Morgan's}] \\
 & = X(Y + Z(X' + Y')(X' + Z'))' \quad [\text{De-Morgan's}] \\
 & = X(Y + Z(X' + X'Y + X'Z + Y'Z))' \\
 & = X(Y + Z(X'(1 + Y + Z) + Y'Z))' \\
 & = X(Y + Z(X' + X'Y'Z))' \quad [\text{properties of 1}] \\
 & = X(X + X'Z) \quad \left\{ \begin{array}{l} a \cdot a = a \\ a' \cdot a = 0 \end{array} \right. \\
 & = XY
 \end{aligned}$$

$$\begin{aligned} &= (A'B) \cdot C \\ &= (A'B)' + C' \\ &= A'B + C' \end{aligned}$$